

Patuakhali Science and Technology University

5th Semester (Level-3, Semester-I) Final Examination - C.P. Sc. Engg. (CSE) (January-June 2018)

Course Code: CIT-313 Course Title: **Computer Organization and Architecture**

Credit Hour: 3.00 Session: 2015-16 Full Marks: 70 Duration: 3 Hours

[Figure in the right margin indicates full marks. Split answering of any question is not recommended.]

Answer any 5 of the following questions. Answer must be brief, relevant and neat.

1. a) A hypothetical machine has 6 instructions

1010=Load AC from memory	1011=Sub memory from AC
0011=Store AC to memory	0110=Store AC to I/O
0101=Add to AC from memory	1101=Load AC from I/O

In this case, the 12-bit address identifies a particular I/O device. Show the program execution (using 16 bit instruction format where prefix 4 bits for opcode) for the following program.

- Load AC from device 99.
- Add contents of memory location 555
- Store AC to memory location 666.

Assume that the next value retrieved from ^{device} memory location 99 is 2 and location 555 contains a value of 3.

- Distinguish between computer architecture and organization.
 - Mention the computer generation with its corresponding technology. What general roles are performed by processor registers?
- Why do we need to study computer organization and architecture?

- Explain the distinction between the written sequence and time sequence of an instruction. Illustrate and list the micro-operations to perform addition between two operands from memory.
 - Differentiate between CISC and RISC architectures. What are their typical characteristics? Give some examples of processors of each category.
 - Why error correction is needed in memory system?

- Mention the key features of RAID 0-RAID 6. Pipeline processor is inefficient for branch instruction. Explain the mechanisms to deal the branches in pipelining.
 - Assume a pipeline with 4 stages: fetch instruction (FI), decode instruction and calculate address (DA), fetch operand (FO), and execute (EX). Draw a timing diagram for instruction pipeline operation for a sequence of 9 instructions, in which the fifth instruction is a branch that is taken and in which there is no data dependencies.
 - Mention data types for multimedia and give 2 multimedia instruction of Intel Pentium product.
 - Mention the advantages and disadvantages of variable length addressing.

- Describe the elements of a machine instruction. Assume numbers are represented in 6-bit two's complemented representation. Show calculation of the following.
 - $-A+B$ (Hex)
 - $-33-17$ (Decimal)

- Find out the quotient and remainder by performing two's complement division operation where divisor = -3 and dividend = -7.
 - Illustrate the arithmetic shift operation.
 - Enlist different addressing mode.

- Why Input/output module is used in computer system? Describe the functions of I/O module.
 - Shortly explain peripheral devices with its classification. Distinguish between program driven and interrupt driven I/O.
 - Distinguish between SRAM and DRAM.

- Compare direct, associative and set-associative memory. Explain replacement algorithms of cache memory.
 - Mention the control signals among system module. Explain multiple interrupt with examples.
 - Give the characteristics of computer family.

Patuakhali Science and Technology University

5th Semester (Level-3, Semester-I) Final Examination of B.Sc. Engg. (CSE)-2018 (January-June)

Course Code: CIT-311 Course Title: Microprocessors and Microcomputers

Credit Hour : 3.00 Session: 2015-16 Full Marks: 70 Duration: 3 Hours

[Figure in the right margin indicates full marks. Split answering of any question is not recommended.]
Answer any 5 of the following questions.

- 1 a) Explain 16 bit instruction format of 8086. What will be the CS: IP of physical address FBCDEh in 8086? 2+2
~~b)~~ For an 80486 descriptor that contains a base address of 0100000Fh, and a limit of 001FFh. What starting and ending locations are addressed by this descriptor considering G=0 and G=1? 3
~~c)~~ Describe following instructions with respect to 8086. 4
 i) MOV 43H[SI], BX
 ii) MOV CS:[AX], DI → DI to memory. E
~~d)~~ "CPU actually works on binary digits"- Justify this statement. 3
- 2 a) Why is memory decoding needed in computer system? Describe 3-state buffer basic input interface. 2+2
~~b)~~ Define *n*-bit processor. 8086 is a 16 bit microprocessor, but its memory mapping is 20 bits. How it is possible? 3
~~c)~~ What is descriptor? Describe the access rights byte of descriptor. 2+2
~~d)~~ How does the pointer register work with their corresponding segment registers? 3
- 3 a) Why is DMA used in computer system? Explain trigger of DMA. 2+2
 b) Describe addressing modes of DSP56300. 3
 c) Give the responsibility coprocessor in computer system. Mention 6 processors with its corresponding coprocessors. 2+2
~~d)~~ Explain the priority between a DMA channel and the core. 3
- 4 ~~a)~~ Distinguish between microcontroller and microprocessor. Give the application of microcontroller. 2+2
 b) Give the features of 8031 microcontroller. Enlist the brands of microcontroller. 2+1
~~c)~~ Write short notes with key features of 2+2
 i) Pentium iv
 ii) Core i5
~~d)~~ Explain floating gate transistor with respect to EPROM. 3
- 5 a) Describe the responsibility of memory management unit in computer system. Determine the number of page table entries that are needed for following combinations of virtual address size (n) and page size (P). 2+2

n	P=2 ^P	#PTE
16	4K	
64	16K	

~~b)~~ What is virtual address? What do you mean by pages in virtual memory system? 1+2
~~c)~~ Define handshaking. Distinguish between 8085 and 8086. 2+2
~~d)~~ Explain "how does onionskin operating system provide resource protection". 3
- 6 a) Why is PPI used in computer system? Explain the basic operation of TSR program. 2+2
~~b)~~ Compare among mode 0, mode 1 and mode 2 of 82C55PPI. 3
~~c)~~ How does the page fault handle in virtual memory system? 4
~~d)~~ Describe the role of operating system in virtual memory. 3

Patuakhali Science and Technology University

B.Sc. Engg. (CSE) 5th Semester Level-3, Semester-I, Final Examination, January, 2019

Course code: CIT-315

Course Title: Artificial Intelligence

Credit hour: 3.00 Full marks: 70

[Figures in the right margin indicate full marks.]

Answer any 5 of the following questions. Split Answering is not recommended.

1. a. Define Artificial Intelligence within 4 different paradigms with examples. 3
~~b. What is rationality? Differentiate between human agent and artificial agent in terms of rationality. 1+2~~
~~c. "Do not measure the performance of an agent based on its behaviour. Measure the performance of an agent based on what is wanted in the environment." – justify these statements. 4~~
~~d. Explain the steps that a problem solving agent uses to solve a problem with an example. 4~~
2. a. What are the components of a solution to a well-defined problem? Explain with an example. 3
~~b. Define a game. Differentiate between games and search problems. 1+2~~
~~c. Demonstrate minimax algorithm and alpha-beta pruning with examples. 3+3~~
~~d. Discuss an example of the application of game theory to a real life scenario. 2~~
3. ~~a. Analyse BFS technique in terms of performance measure. Provide reasons to every measure. 3~~
~~b. "Exponential time or space complexity is very expensive" – justify with an example in terms of time and memory requirements. 3~~
~~c. Compare iterative deepening search (IDS) and breadth first search (BFS) in terms of performance measure. Which one is preferred when the search space is large and the depth of the solution is unknown? Explain with an example. 2+2~~
~~d. Consider the following scenario for 8-puzzle game. Figure 3.d.1 is the initial state and figure 3.d.2 is the goal state. Which of the uninformed search technique would be best suited? Justify your answer in terms of performance measure 4~~
4. ~~a. How does greedy best first search work? Show using figure 1. What are its properties? 2+2~~
~~b. Demonstrate how greedy best-first search is incomplete using figure 1. 4~~
~~c. Demonstrate the working principle of A* search to find a path from Oradea to Bucharest (figure 1) 4~~
~~d. Define: Admissible Heuristic, Optimally Efficient. 1+1~~
5. ~~a. Compare supervised and unsupervised learning. Explain the structure of artificial neuron. 2+2~~
~~b. Why fuzzy logic is used in neural network? Explain the neural network with its classification. 2+3~~
~~c. Explain fuzzy inference system. Distinguish between digital computer and neural networks. 3+2~~
6. ~~a. Explain member function with its features. Distinguish between crisp set and fuzzy set. 3+2~~
~~b. Define defuzzification. Explain the importance of neural network with respect to artificial intelligence. 1+3~~
~~c. Consider the following linguistic variables and membership functions (figure 6.c.1) with respect to fuzzy logic. Calculate the speed of driving at 65% Cloud Cover and 37 F° 5~~

8		6
5	4	7
2	3	1

Figure 3.d.1

	1	2
3	4	5
6	7	8

Figure 3.d.2

Rules	Linguistic-variables
i) If it's sunny and cool drive fast	i) Temp: {freezing, cool, warm, hot} (Fig.: i)
ii) If it's overcast and freezing drive slow	ii) Cloud Cover: {overcast, partly cloudy, sunny} (Fig.: ii)
	iii) Speed: {slow, fast} (Fig.: iii)

$$f(n) = g(n) + h(n)$$

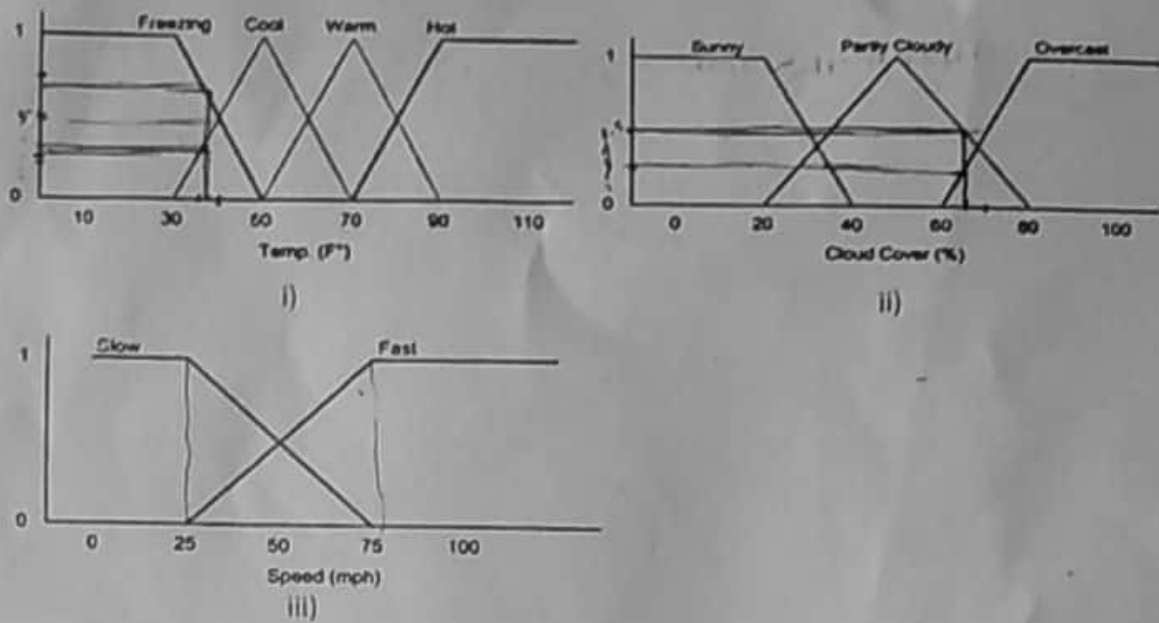


Figure 6.c.1

Romania with step costs in km



Figure 1

[Figure in the right margin indicates full marks. Split answering of any question is not recommended.]
 Answer any 5 of the following questions.

- 1 a) The following is a dump of a TCP header in hexadecimal format. 7
05320017 07000001 00000000 500207FF 00000000
 a. What is the source port number?
 b. What is the destination port number?
 c. What is the sequence number?
 d. What is the acknowledgment number?
 e. What is the length of the header?
 f. What is the type of the segment?
 g. What is the window size?
- Source Port: 1058
 Destination Port: 80
 Sequence: 1058
 Acknowledgment: 80
 Header Length: 20
 Window Size: 65535
 Offset: 0
 Reserved: 0
 Control Bits: 00000000
 Unused: 0
- b) An ISP is granted a block of addresses starting with 120.60.4.0/22. The ISP wants to distribute these blocks to 100 organizations with each organization receiving just eight addresses. Design the subblocks and give the slash notation for each subblock. Find out how many addresses are still available after these allocations. 5
- c) Why do we need a DNS system when we can directly use an IP address? 2
- 2 a) What is NAT? How can NAT help in address depletion? 3
 b) Briefly define subnetting and supernetting. How do the subnet mask and supernet mask differ from a default mask in classful addressing? 5
 c) A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28. Find out the first address, last address, and the number of addresses in the block. Also show the network configuration for the block. 6
- 3 a) What is CIDR? Find the netid and the hostid of the following IP addresses: 5
 i. 217.34.38.0
 ii. 211.23.1.6
 iii. 114.56.7.83
 iv. 207.24.12.0
- b) In TCP, if the value of HLEN is 1000, how many bytes of option are included in the segment? 2
- c) A TCP connection is using a window size of 12,000 bytes, and the previous acknowledgement number was 22,001. It receives a segment with acknowledgement number 24,001 and window size advertisement of 12,000. Draw a diagram to show the situation of the window before and after. 5
- 4 a) In the standard Ethernet, if the maximum propagation time is 25.6 μ s, what is the minimum size of the frame? 2
- b) Illustrate the *Ethernet Frame*. Also show the minimum and maximum length. 4
- c) Write short notes on:
 i. Wireless LAN
 ii. Point-to-point WAN
 iii. Switched WAN 6
- d) How are options negotiated in TELNET? 4
- 5 a) Show a request that retrieves the document /usr/users/doc/doc1. Use at least two general headers, two request headers, and one entity header. Also show the response for a successful request. 5
- b) Write down the differences among repeaters, bridges, and routers. 4
- c) What is the purpose of the inverse domain? 2
- d) Define three types of web documents? Active, static, dynamic. 3
- 6 a) When both sender and receiver are connected to the mail server via a LAN or a WAN, what types of agents are needed and how many? Describe the situation. 7
- b) Differentiate between circuit switching and packet switching. 3

Final Examination of B.Sc. Engineering in CSE Level: 3 Semester: I Session: 2015-16

Course Code
CCE 311

Course Title
Numerical Methods

January-June
2018

Credit: 03
Time: 03 Hr
Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

Put appropriate figure/example if necessary

1. A Determine the real roots of $f(x) = -0.4x^2 + 2.2x + 4.7$ 2+4
 i. Using the quadratic formula
 ii. Using three iterations of the bisection method to determine the highest root. Employ initial guesses of lower $x_1 = 5$ and $x_4 = 10$. Compute the estimated error e_a and true error e_t after each iteration.

1. ~~A~~ Explain accuracy and precision with example. Write down different types of ^{error} in numerical analysis with the formula to calculate them. 2+3

1. ~~C~~ Derive the formula of false position method with proper figure. 2+3

2. ~~A~~ How do you use Taylor series in the derivation of Newton-Raphson method? Explain the pitfalls of Netwan-Raphson method with example. 3+3

2. ~~B~~ Use simple fixed-point iteration to locate the root of $f(x) = -0.9x^2 + 1.7x + 2.5$ 5
 Use an initial guess of $x_0 = 5$ and iterate until $e_a \leq 0.01\%$

2. ~~A~~ State the differences between secant and false-position method. 3

3. ~~A~~ Use Mullar's method with guesses of x_0, x_1 and $x_2 = 4.5, 3.5$ and 5.5 respectively, to determine a root of the equation $f(x) = x^3 - 13x - 12$ 6
 Note that the roots of this equation are -3, -1, and 4

3. ~~B~~ Apply Cholesky decomposition to the symmetric matrix 5

$$[A] = \begin{bmatrix} 6 & 15 & 35 \\ 15 & 55 & 225 \\ 35 & 225 & 979 \end{bmatrix}$$

3. ~~C~~ Write short notes on 1.5+1.5
 i. LU decomposition
 ii. Lower and Upper triangular matrix

4. ~~A~~ Fit a second-order polynomial to the data of the following table. Also find out standard error S_{yx} 7

x_i	y_i
1	7.7
2	13.6
3	27.2

4. B Use Simpson's 3/8 rule to integrate $f(x) = 0.2 + 25x - 200x^2 + 675x^3 - 900x^4 + 400x^5$ from $a=0$ to $b=0.8$ 7
 Use it in conjunction with Simpson's 1/3 rule to integrate the same function for five segments
 Find out E_t for both cases. 7

5. A Derive equation for linear regression and show criteria for a "Best" fit. $y = a_0 + a_1x$ 7

5. B Given $dy/dx = 1/2(x+y)$, $y(0)=2$, $y(0.5)=2.636$, $y(1.0)=3.595$, $y(1.5)=4.968$. Find $y(2)$ by Milne's method. 7

6. A Use the point-slope method to numerically integrate $dy/dx = -2X^3 + 12X^2 - 20X + 8.5$ from $X=0$ to $X=4.0$ with a step size 0.5. The initial condition at $X=0$ is $Y=1$. 7

6. B Given $dy/dx = 1+xy$ and $y(0)=1$. Calculate $y(0.1)$, $y(0.2)$ using Picard's method. 7